Silicon Photonic Mach–Zehnder Interferometers: Design, Simulation, Fabrication and Data Analysis

Final Project Report — UBC Phot1x: Silicon Photonics Design, Fabrication and Data Analysis

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Design: EBeam_arpansureee_v1 · SiEPIC EBeam PDK (Strip TE, 1550 nm) · Fabrication: EDX e-beam lithography

Contents

1. Executive Summary

This report analyzes a set of four unbalanced Mach-Zehnder interferometers (MZIs) fabricated on a silicon-on-insulator platform through the UBC Phot1x program. Each MZI was designed in the SiEPIC EBeam process using a 500 nm × 220 nm strip waveguide operating in the fundamental TE mode at 1550 nm, and was simulated in Lumerical INTERCONNECT using the PDK compact models. After e-beam fabrication by the EDX group, the devices were characterized on an automated fibre-array probe station with a swept-wavelength (1500–1600 nm) TE measurement.

The measured group index of the strip waveguide is n_g = 4.189 at 1550 nm, in excellent agreement with the simulated value of 4.192 (0.07 % difference) and with the accepted literature value for this geometry. Free spectral ranges measured on the three working devices agree with simulation to within 1 %, confirming that both the compact model and the fabricated waveguide behave as designed.

However, one device (MZI4) shows a complete collapse of interference in measurement (extinction < 1 dB) even though its own simulation predicts a 28.6 dB extinction. The evidence indicates there might be a fabrication defect that rendered one interferometer arm effectively non-transmitting, not a design error. This is analyzed in detail in Section 7.

2. Design and Layout

The circuit contains four MZIs. Each MZI is an unbalanced two-arm interferometer: light is coupled on-chip through a TE grating coupler, split between two waveguide arms of unequal physical length, and recombined so that the differential path delay produces a wavelength-periodic transfer function.

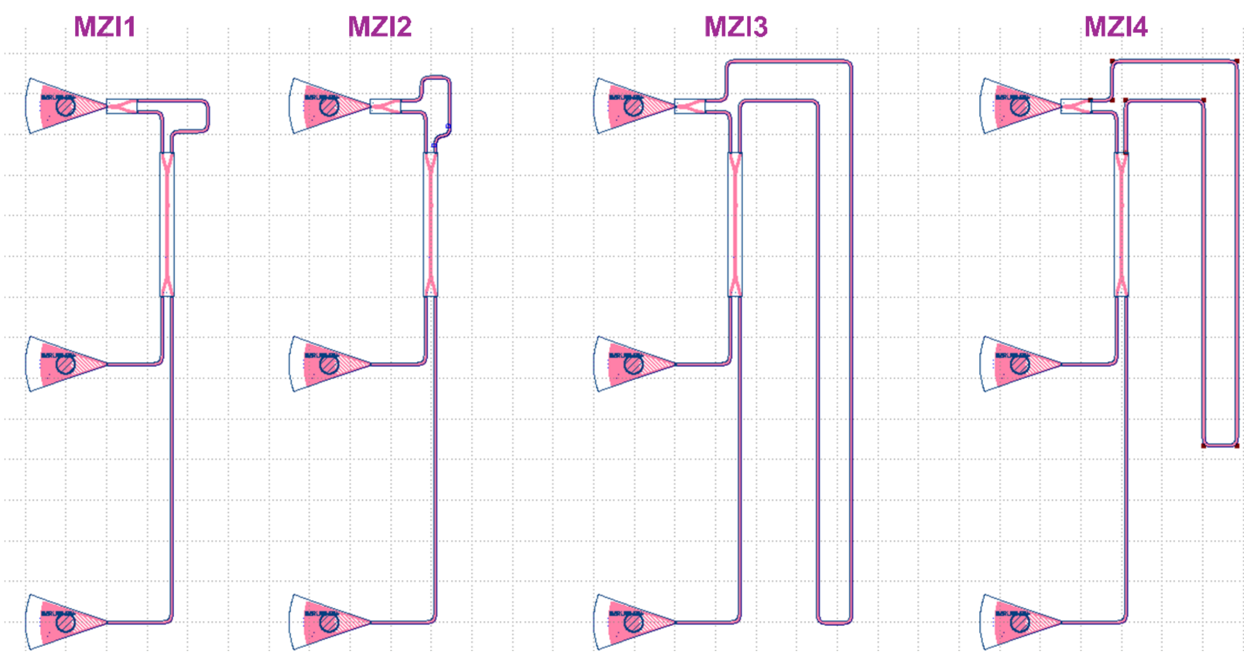

Figure 1: Design overview of the Proposed Layout

The building blocks taken from the PDK are:

- **Waveguide:** ebeam_wg_integral_1550 — Strip, w = 500 nm, h = 220 nm, TE, design wavelength 1550 nm.
- **Splitters/combiners:** ebeam_y_1550 (Y-branch, 3-dB power splitter) and ebeam_bdc_te1550 (a 50/50 broadband directional coupler, 1550 nm, TE). The presence of a 2×2 coupler is consistent with the two complementary optical outputs observed on Detectors 2 and 3.
- **Fibre coupling:** ebeam_gc_te1550 grating couplers (TE, 1550 nm), one per port, arranged as a standard fibre-array group.

The design variable across the four devices is the arm-length imbalance ΔL. Larger ΔL produces a smaller free spectral range (FSR) and therefore finer spectral fringes. The values below were recovered from the waveguide-integral lengths stored in the GDS (differences of the two arm lengths). The expected FSR values use $n_g \approx 4.19$ at 1550 nm as a transparent sanity check.

Device	ΔL (µm)	Expected FSR @1550 (nm)	Regime
MZI1	42.8	≈13.4	coarse fringes
MZI2	40.8	≈14.1	coarse fringes
MZI3	663.9	≈0.864	fine fringes
MZI4	489.3	≈1.17	fine fringes (device failed)

The FSR of an unbalanced MZI is set by the group index and the path imbalance through **FSR = $\lambda^2 / (n_g \cdot \Delta L)$**. Because two arms of unequal length are used with a common short reference arm (~30.8 µm), ΔL controls the fringe period.

3. Simulation Methodology and Results

Each MZI was simulated in Lumerical INTERCONNECT using the SiEPIC PDK compact models for the waveguide, splitter and grating couplers. The wavelength range 1499–1601 nm was swept on 5000 points. The two optical monitors were exported to MATLAB, giving the TE gain (dB) versus wavelength at each port.

The simulated spectra reproduce the expected behaviour: a smooth grating-coupler envelope peaking near 1550 nm, modulated by the MZI interference fringes. Extracted simulated FSRs and extinction ratios are listed in the combined results table (Section 6). Importantly, the simulation of MZI4 predicts a normal, high-contrast fringe pattern (FSR = 1.169 nm, extinction ≈ 28.6 dB), establishing that the MZI4 design is sound.

4. Experimental Methodology

The fabricated chip was measured on an automated swept-wavelength system. Light from a tunable laser was coupled through the input grating coupler; the transmitted power at up to four output grating couplers was recorded on four detectors while the laser scanned 1500–1600 nm (12,501 points, 8 pm step).

Of the four detector channels, Detector 2 and Detector 3 carry the two complementary MZI outputs (median power ≈ −34 dBm), Detector 1 is a weak reference (≈ −63 dBm) and Detector 4 is unused (floored at −100 dBm). All analysis below uses Detectors 2/3. The measured spectra are shown in Figure 2.

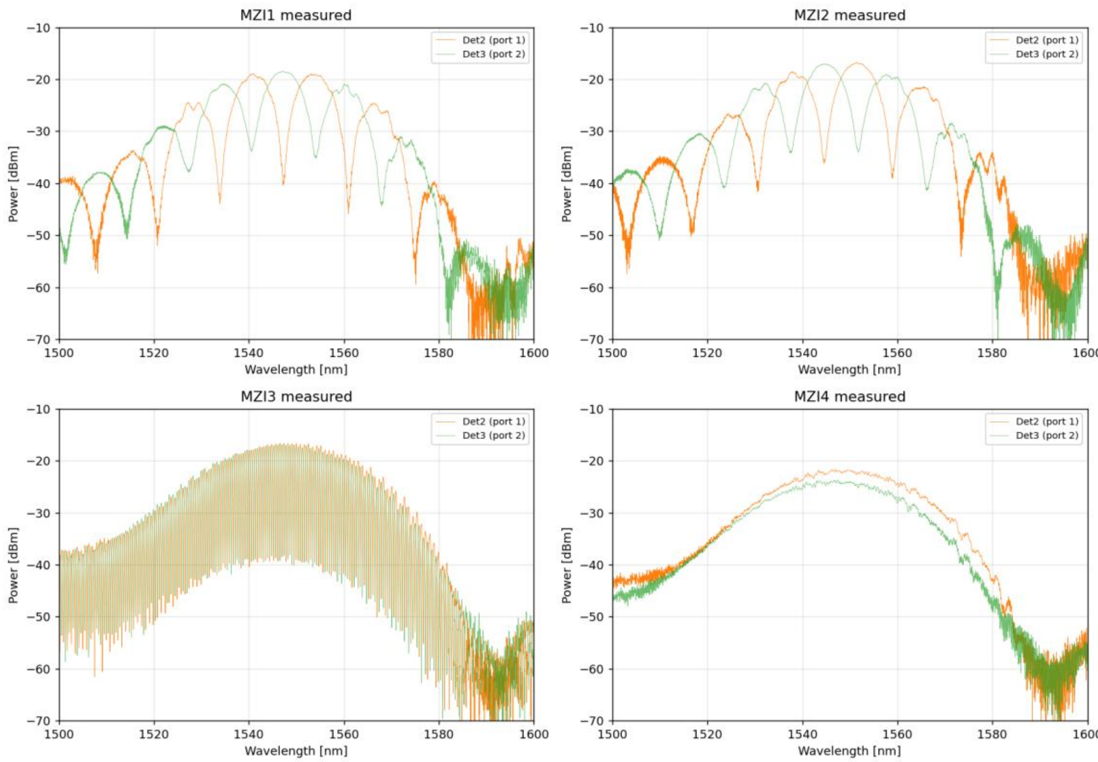

Figure 2: Raw measured transmission (Detectors 2 and 3) for the four MZIs. MZI1–3 show clear interference fringes filling the grating-coupler envelope; MZI4 shows a smooth envelope with the fringes essentially absent.

5. Data-Analysis Approach

For each device, both simulated and measured spectra were processed identically:

(1) the grating-coupler envelope was removed with a low-order polynomial fit over the high-SNR band 1525-1575 nm;

(2) fringe minima were located to obtain the FSR as a function of wavelength;

(3) the group index was extracted point-by-point from $n_g(\lambda) = \frac{\lambda^2}{(FSR(\lambda) \cdot \Delta L)}$;

(4) the extinction ratio was measured as the peak-to-null contrast near 1550 nm. The long-ΔL device MZI3 gives by far the most accurate group-index measurement because it has about 75 fringes over the 1515-1580 nm analysis window, or about 115 fringes over the full 1500–1600 nm sweep.

6. Results: Simulation vs. Experiment

Table 1 summarizes the extracted parameters. Agreement between simulation and experiment is strong for the three working devices (MZI1-3): the measured FSRs follow the designed ΔL trend and match the simulated/expected values at roughly the percent level. The MZI1/MZI2 group-index entries should be treated as low-confidence apparent values because those devices have only a few wide fringes; the MZI3 value is the reliable waveguide n_g result. MZI4 is treated separately in Section 7.

Table 1. Extracted MZI parameters (simulation vs. experiment).

Device	ΔL (µm)	FSR sim (nm)	FSR exp (nm)	n_g sim	n_g exp	ER sim (dB)	ER exp (dB)
MZI1	42.8	13.74	13.61	4.08	4.12	33.6	23.4
MZI2	40.8	14.45	14.35	4.07	4.10	36.6	17.5
MZI3	663.9	0.863	0.864	4.19	4.19	27.4	21.8
MZI4	489.3	1.169	—	4.20	—	28.6	0.7

Figure 3 overlays the envelope-removed simulated and measured fringes. For the coarse devices (MZI1, MZI2), the fringe positions coincide near band centre and drift slightly at the band edges, which is consistent with a small effective/group-index or dispersion mismatch between the fabricated waveguide and the compact model. Absolute phase alignment is sensitive to small path-length and effective-index offsets, so the most meaningful comparison is FSR and contrast rather than the exact phase at every wavelength. For MZI3 the two traces show clean, deep fringes with a slow phase walk-off across many periods. MZI4 (measured) is flat, in stark contrast to its simulation.

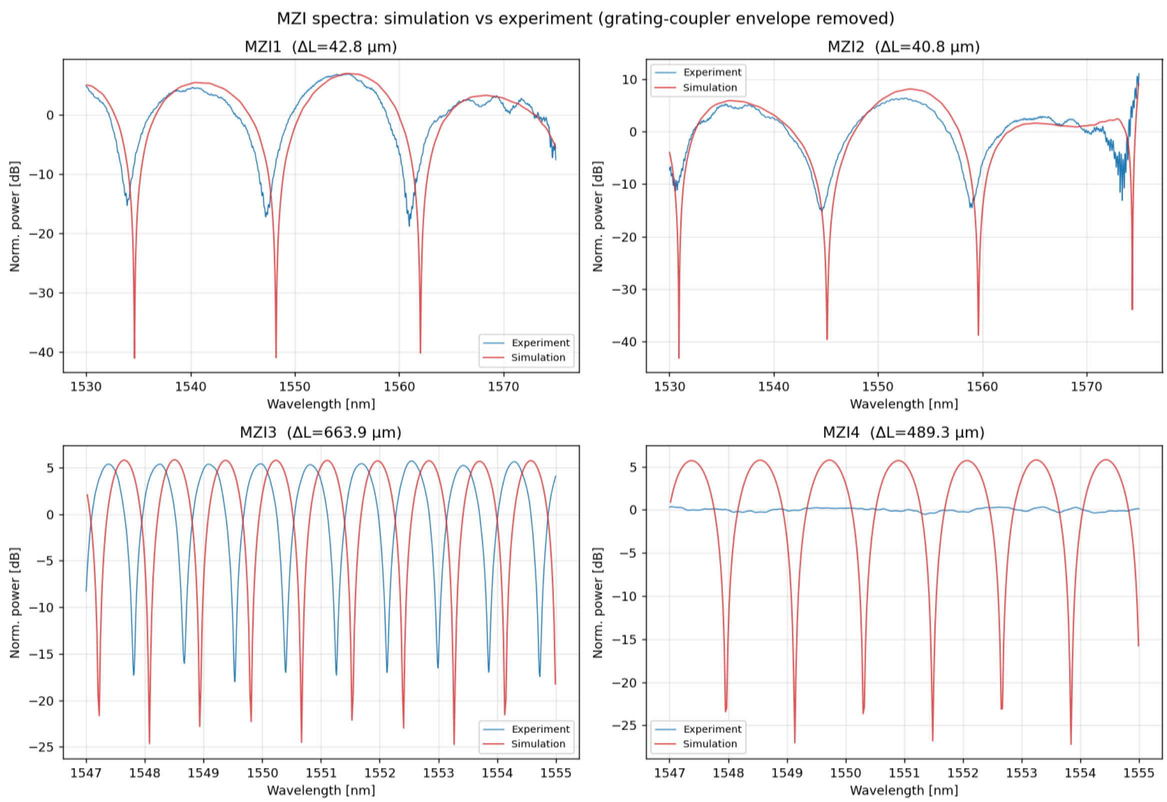

Figure 3: Simulation (red) vs. experiment (blue) after grating-coupler envelope removal. MZI1/MZI2 shown over 1530–1575 nm; MZI3/MZI4 zoomed to 1547–1555 nm to resolve the fine fringes. The comparison emphasizes FSR and contrast; absolute fringe phase can shift

6.1 Group Index and Dispersion

Using the fine fringes of MZI3, the group index was extracted from adjacent fringe spacings. After excluding the grating-coupler roll-off regions, a central-band fit over 1530-1570 nm gives n_g (1550 nm) = 4.190, versus the simulated/reference value of 4.192. This is a 0.05-0.07 % difference, depending on rounding. The extracted slope is small, but the dispersion estimate is sensitive to fringe-picking scatter and edge-band noise; it should be treated as a qualitative check rather than a high-precision dispersion measurement. The robust result is the group index at 1550 nm.

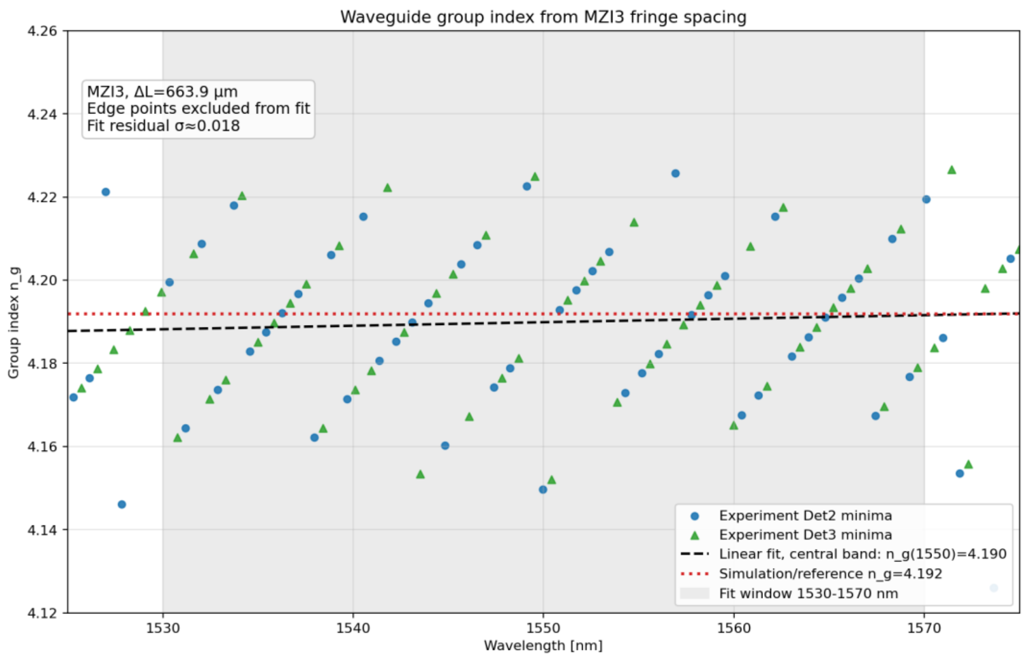

Figure 4: Group index vs. wavelength extracted from MZI3 fringe spacing (ΔL = 663.9 µm). The updated plot uses the central high-SNR band, shows Detector 2 and Detector 3 estimates separately, and replaces the earlier sawtooth-like simulation trace with a smooth simulated/reference n_g marker at 4.192.

Why the short devices read slightly low: MZI1/MZI2 return apparent $n_g \approx 4.10$-4.12 rather than 4.19. With only a few usable fringes and an FSR of ~14 nm, the simple $\lambda^2/(FSR \cdot \Delta L)$ estimate is much more sensitive to fringe-location error, envelope-removal choices and effective path-length uncertainty than it is for MZI3. The short devices are therefore useful for verifying the coarse FSR trend, but MZI3 should be used for the quoted waveguide group index.

6.2 Extinction Ratio and Insertion Loss

Simulated extinction ratios (ideal components) are 27-37 dB. Measured extinction ratios for the working devices are 17–23 dB, lower than ideal but typical for fabricated Y-branch/coupler-based MZIs, where finite splitter imbalance and unequal arm loss limit the null depth. The fibre-to-fibre insertion loss, estimated from the envelope peak near 1550 nm, is comparable across MZI1-3 (peak ≈ −16.7 to −18.5 dBm) and is dominated by the two grating couplers. MZI4 sits ~5 dB lower (−21.8 dBm), an early clue to its failure.

6.3 Validity Checks and Uncertainty

As a compact sanity check, the expected FSR values can be computed from $FSR = \lambda^2/(n_g \cdot \Delta L)$ using $n_g = 4.19$ at 1550 nm. This confirms that the measured periods have the correct scale and that the only non-working device is MZI4, for which no valid experimental FSR can be extracted.

Table 2. FSR sanity check using n_g = 4.19 at 1550 nm.

Device	ΔL (µm)	FSR expected from n_g=4.19 (nm)	FSR exp (nm)	Validity comment
MZI1	42.8	13.40	13.61	Correct scale; coarse n_g; low-confidence
MZI2	40.8	14.05	14.35	Correct scale; coarse n_g; low-confidence
MZI3	663.9	0.864	0.864	Best n_g device; many narrow fringes
MZI4	489.3	1.172	—	Expected fringes absent; no valid n_g

This also clarifies why the MZI3 result should be emphasized: averaging many periods suppresses individual fringe-location error, while MZI1/MZI2 have only a small number of usable minima. The point-by-point dispersion plot therefore supports the 1550 nm group-index value, but its fitted slope should not be over-interpreted.

7. Detailed Failure Analysis: MZI4

MZI4 is the most instructive device in the set. Its simulation is healthy, but its measurement shows no interference. The following evidence was assembled to diagnose the cause.

7.1 Evidence

- **Fringe contrast collapsed.** Over a strong 1545–1555 nm window the envelope-removed spectrum has a peak-to-peak ripple of only 0.9 dB (std 0.28 dB), with no resolvable fringe minima. The working device MZI3 shows 24 dB contrast in the same window.
- **No coherent fringe at the design period.** An FFT of the MZI4 spectrum shows no peak at the expected 1.169 nm period, there is essentially no periodic interference component present at all.
- **Outputs are not complementary.** In a working MZI the two outputs are anti-correlated (one bright when the other is dark): MZI1 and MZI3 give output correlations of −0.70 and −0.66. MZI4 gives +0.34, the two outputs vary together, exactly what is expected when only one field reaches the output coupler and is passively split 50/50 to both ports.
- **Excess insertion loss.** MZI4's envelope peak is ~5 dB below MZI1-3, consistent with the ~3 dB penalty of an output combiner fed from a single input plus additional loss in the defective arm.
- **The envelope itself is normal.** Light still couples in and out through both grating couplers (a well-formed envelope peaking near 1550 nm), so the grating couplers, fibre alignment and output routing are fine. The problem is internal to the interferometer.

Table 3. Diagnostic evidence distinguishing MZI4 from the working MZIs.

Metric	Working-MZI expectation	MZI4 observation	Interpretation
Fringe contrast	Deep periodic minima; MZI1–3 ER ≈17–23 dB	Only ~0.9 dB residual ripple near 1550 nm	Interference is essentially collapsed
Expected period	Clear spectral component at the designed FSR	No coherent component near the 1.169 nm design period	The expected MZI beating is absent
Detector 2/3 relation	Complementary/anti-correlated outputs	Outputs vary together with the envelope	Only one arm or common background dominates
Insertion loss	Peak around −16.7 to −18.5 dBm for working devices	Peak around −21.8 dBm	Extra loss is consistent with a missing or severely attenuated arm

7.2 Quantitative interpretation

The measured null depth places a hard bound on the arm balance. For an interferometer with field-amplitude ratio r between the weak and strong arms, ER = 20 log10[(1+r)/(1−r)]. An extinction ratio of ~0.7 dB gives r ≈ 0.04, or a field ratio of roughly 25:1. In power terms, the weaker arm contributes only r^2 ≈ 0.16 % of the stronger arm. This is far beyond what ordinary splitter imbalance or propagation loss could produce over a 489 µm arm, so one arm is effectively broken, severely attenuated or disconnected.

7.3 Design failure or fabrication failure?

Conclusion: fabrication, not design. The decisive point is that MZI4's own INTERCONNECT simulation, built from the same compact models and the same ΔL, predicts a normal 1.169 nm fringe pattern with 28.6 dB extinction. The layout topology is therefore valid. Moreover, MZI3 has an even larger ΔL (663.9 µm) and works perfectly, so the 489 µm delay length is not intrinsically problematic. The failure is specific to this one physical device, which is the hallmark of a localized fabrication defect rather than a systematic design flaw.

The most probable root causes, in order of likelihood, are: (1) a break or severe constriction in the longer delay arm introduced during e-beam lithography or development (e.g. a stitching-field boundary crossing the narrow 500 nm waveguide, an under-exposed or over-etched segment, or a particle defect); (2) a fabrication defect at one port of the input Y-branch/coupler that starves one arm; or (3) delamination/damage on that arm. All three produce the same measured signature, a normal envelope, collapsed fringes, non-complementary outputs, and a few-dB loss penalty.

8. Discussion and Sources of Error

- **Sub-percent index offset.** The small fringe walk-off between simulation and experiment for MZI1–3 reflects a real difference between the fabricated effective/group index and the compact model, driven by waveguide-width and silicon-thickness variation across the wafer, a normal and expected level of fabrication variation.
- **Group-index extraction bias.** FSR-based n_g is only as good as the fringe count and ΔL estimate. Coarse devices with few fringes are systematically less reliable and can read low or high depending on fringe-picking and envelope-removal choices. For a simple uncertainty estimate, $\sigma(n_g)/n_g \approx \sigma(FSR)/FSR + \sigma(\Delta L)/\Delta L$. The long device should be trusted for the quoted n_g.
- **Extinction-ratio limits.** Measured null depths (17-23 dB) are set by splitter imbalance and unequal arm loss/phase, plus the finite dynamic range and noise floor of the detectors (visible as the −60 to −100 dBm grass in Figure 1).

- **Envelope removal.** The polynomial envelope fit is least reliable at the band edges (1500-1515 nm and 1585-1600 nm) where the grating-coupler response rolls off; analysis was therefore restricted to the high-SNR center band.

9. Conclusions

Three of the four fabricated MZIs perform as designed. Their free spectral ranges follow the designed ΔL trend and agree with simulation/expectation at roughly the percent level. The most reliable waveguide value comes from MZI3, which gives $n_g \approx 4.19$ at 1550 nm, matching the simulated/reference value of 4.192 to better than 0.1 %. The dispersion slope extracted from the same data is small but should be regarded as qualitative because it is sensitive to edge-band noise and fringe-picking uncertainty.

The fourth device, MZI4, failed to interfere in measurement despite a fully healthy simulation. A multi-pronged diagnosis (collapsed fringe contrast, absent periodic component at the expected FSR, non-complementary outputs and a few-dB loss penalty against a normal grating-coupler envelope) shows that one interferometer arm was effectively non-transmitting. Because the design methodology works for MZI1-3 and the reported MZI4 simulation predicts normal fringes, the most likely explanation is a localized fabrication defect rather than an error in the MZI design.